A Report on the Flight of Delta II's Redundant Inertial Flight Control Assembly (RIFCA)

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INTRODUCTION

AlliedSignal's Redundant Inertial Flight Control Assembly (RIFCA) is a practical, cost-effective, fully fault tolerant inertial system currently used on the Boeing Delta II launch vehicle. The RIFCA program is now in the production phase, providing flight hardware for the Delta II program, and has successfully flown on 17 Delta II launches to date. RIFCA measures vehicle inertial angular rates and linear accelerations, and provides for the computational processing of guidance, navigation, flight controls and mission sequencing data.

This paper describes key elements of the RIFCA top-level system design, including redundancy concepts, hardware and software, all integrated to provide a fail-op capability. In addition, it summarizes RIFCA system level performance capability and Boeing Delta II launch vehicle mission accuracy based on actual system/mission launch data. This paper also discusses planned enhancements to RIFCA.

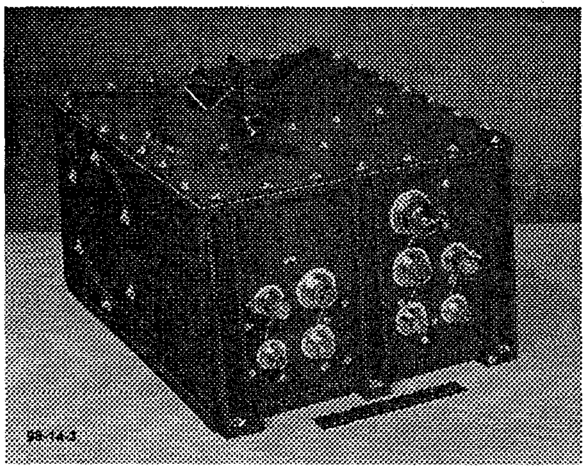

Figure 1: AlliedSignal Redundant Inertial Flight Control Assembly (RIFCA)

Table 1: RIFCA Characteristics

Instrument Complement	6 RL-20 Gyros 6 QA-3000-010 Accelerometers
Rate Capability	± 400 deg/sec All Axes Specified: ± 30 deg/sec
Acceleration Capability	± 16.5 g Spec: ± 15 g Along Centerline ± 5 g Perpendicular to Centerline
Processor	MIL-STD-1750A Architecture (BX-1750)
Memory	64K Static RAM 16K Bootstrap PROM 64K Program Store (EE PROM)
Electrical Interface	Dual MIL-STD-1553 Bus Controllers RS-422 Data Link (GSE)
Construction	Aluminum Investment Casting
Volume	17 x 13.5 x 9.3 in
Weight	73 lbs Typical
Power	73 Watts Typical
Reliability	0.9999986; Spec: 0.99999

HARDWARE DESIGN OVERVIEW

RIFCA (pictured in Figure 1 with key characteristics summarized in Table 1) measures vehicle inertial angular rates and linear accelerations, and provides for the computational processing of guidance, navigation, flight controls and mission sequencing data. RIFCA distributes control and sequencing commands to vehicle interface hardware via redundant 1553 data buses. This is accomplished in a single box design using a three-lane approach as shown in Figure 2.

Specifically, RIFCA consists of an Inertial Sensor Assembly (ISA) and an Inertial Processing Electronics (IPE) section. The ISA contains six AlliedSignal RL-20 Ring Laser Gyros (RLGs), six AlliedSignal QA3000 pendulous accelerometers, and six high voltage power supplies. A RIFCA RLG and accelerometer are shown in Figure 3. All of the ISA components are contained in individual pressurized cavities. The inertial sensors are orientated as two

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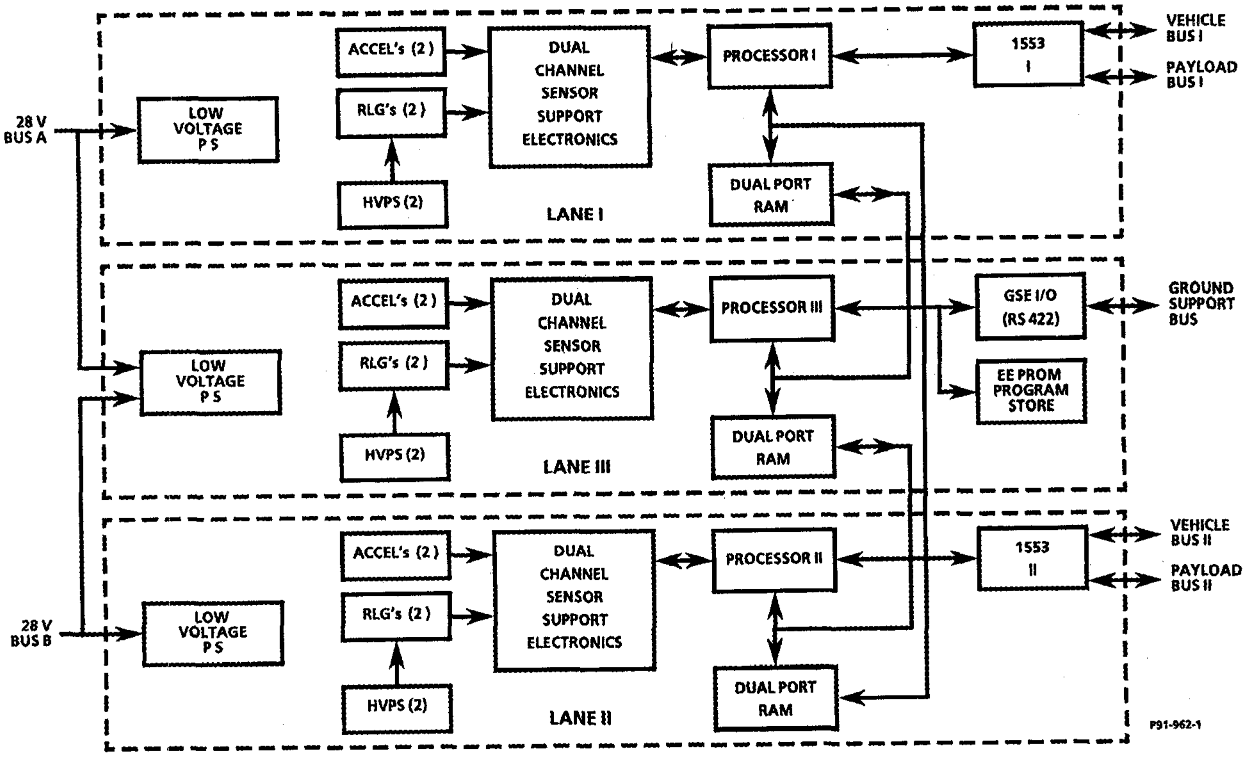

Figure 2: RIFCA Triple Modular Redundant (TMR) Architecture

orthogonal triads, skewed to each other and to the vehicle frame, with two RLGs and two accelerometers assigned to each of RIFCA's three lanes. The IPE contains three MIL-STD-1750A computers based on the AlliedSignal BX-1750A microprocessor, two MIL-STD-1553 communication ports to interface to the vehicle's redundant power and control subsystems and telemetry, and an RS-422 serial port for ground communications.

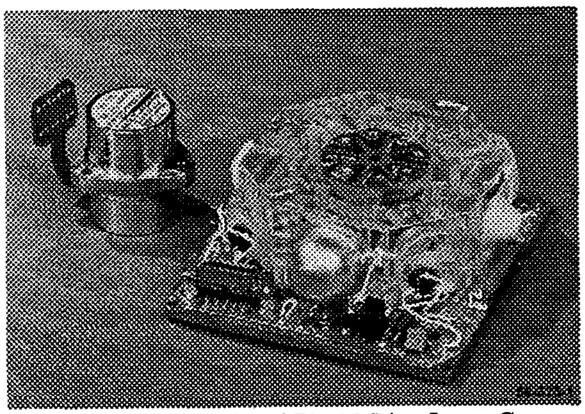

Figure 3: AlliedSignal RL-20 Ring Laser Gyro and QA3000 Pendulous Accelerometer

The Triple Modular Redundant (TMR) architecture shown in Figure 2 ensures that there are no single point failures in the design and provides sufficiently improved reliability to move away from the traditional, high cost utilization of Class S devices. The TMR design consists of three identical sensor/processing channels with a minimum of cross strapping. Inertial sensor data is shared between lanes via dual port RAMs. Hardware voting is implemented for real time clock and output data. Vehicle and payload bus controller redundancy is provided via the 1553 serial communication links in Lanes I and II, each of which broadcast identical time synchronized data.

The center lane (Lane III) controls ground support communication including memory load/store and program control/debug, and accesses Lanes I and II via the dual port RAMs. The flight program and sensor calibration constants are located in the non-volatile program store EEPROM. The flight program is loaded from the ground into Lane III, then distributed from the EEPROM to the three processors' working memory via the dual port RAM interface. The ground checkout of the vehicle uses the contents stored in the EEPROM, thereby preventing inadvertent programming errors in the final launch version. The calibration constants are located in a secured EEPROM partition accessible at the factory. Two 28 Volt buses provide power to the RIFCA system, where the center lane operates from either power source.

Inertial Sensor Data

The six RLGs are mechanically dithered to prevent lock-in, with three nominal dither frequencies being used (one for each lane). The data from these instruments is collected as up and down counts over their respective dither periods. This is done by transferring the contents of continuous up and down sensor counting registers to holding registers at fixed zero crossover points in the dither cycle. The data is then read asynchronously into the processor from the holding registers after each real time interrupt, which occurs every 2.5 milliseconds.

Accelerometer data is obtained by taking an analog output and converting to up and down counts via a Voltage to Frequency (V/F) converter. These counts are continuously summed and transferred to holding registers synchronously with the 2.5 millisecond data read cycle.

The gyro data collection scheme and accelerometer V/F converters provide the integral of sensed rate and acceleration. Data resolution is defined by the pulse weight of the sensors, which is nominally 0.65 arcsec/pulse for the gyros and 0.01 ft/sec/pulse for the accelerometers.

Processor Operation

Each lane's processor compensates its inertial sensor data for scale factor and bias, both determined from system level calibration. This data is then shared with other lanes via the dual port RAMs, and then system level compensations of coning, sculling, misalignment and pre-launch bias updates are applied to the data in each lane. Processing continues in each lane with gyro and accelerometer sensor redundancy algorithms, followed by alignment and navigation algorithms. At this point all three processors are solving the computation algorithms independently and sharing a sample of their results through the dual port RAMs. This shared data is the basis used in each processor to vote for a defined output.

Coordinate Frame Definitions

The vehicle body frame is defined in Figure 4 as X up, Z along the azimuth vector (which is defined as positive clockwise from North) and Y creating a right handed coordinate system. The RIFCA frame is defined to be coincident with the body frame, with the Y and Z axes rotated about the X body axis by a fixed angle. The sensor frame is defined by the physical orientation of each sensor's input axis, and is nominally skewed relative to the RIFCA frame.

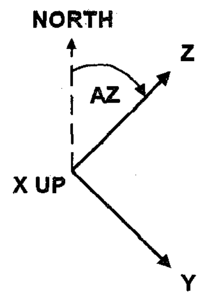

Figure 4: Body Frame Definition

System level calibration alignment data defines the sensor to RIFCA coordinate transformation matrices for three possible sensor combinations (all lane combinations taking two lanes at a time). These matrices are adjusted in the program start routine to account for the orientation of RIFCA relative to the body frame using a fixed rotation matrix. The sensor redundancy algorithms are valid only if the body and RIFCA X axis coincide, as is the case when RIFCA is in the vehicle, on the pad and in flight conditions.

Sensor Redundancy

Each lane's processor implements both gyro and accelerometer sensor redundancy algorithms using shared sensor data. In order to provide sensor failure detection and isolation capability, the sensor data is transformed into the body frame using three different combinations of the six sensors. Each computed body frame triad consists of sensor data from only two of the three lanes. By definition, Lane I contains sensors 1 and 4, Lane II contains sensors 2 and 5, and Lane III contains sensors 3 and 6. Based on these definitions, the three computed triads are defined as follows: Triad 1 is generated from Lanes I and II (sensors 1, 4, 3 and 6), Triad 2 from Lanes I and III (sensors 1, 4, 2 and 5), and Triad 3 from Lanes II and III (sensors 2, 5, 3 and 6).

Both failure detection (by exceeding a threshold) and failure isolation (by the largest of the parameters) can be achieved by observing the integrated difference in the X axis components of this data in three separate parameters, each amplifying a different lane's errors. Isolation consists of selecting the sensor triad which is not contaminated by corrupt lane data, as summarized in Table 2. No significant degradation to navigation occurs if one or all of the processors trigger sensor redundancy logic since each of the computed triads can meet mission objectives. The average of the three triads is used for navigation and control data if no failure is detected. The design provides for independent accelerometer and gyro redundancy, which allow for an accelerometer and a gyro failure in different lanes.

Table 2: Sensor Redundancy Failure Isolation

Lane or Sensor Failure	Selected Triad
None	Average
Lane I (Sensor 1 and/or 4)	3
Lane II (Sensor 2 and/or 5)	1
Lane III (Sensor 3 and/or 6)	2

Figure 5 shows a typical example of how the gyro sensor redundancy logic works during a Lane I power down test. When all of the failure detection parameters (redundancy terms) remain less than the threshold, no failure is detected, and the average of the three triads is selected. When the redundancy terms exceed the threshold, a failure is detected and the appropriate triad of data is selected based on the largest term. In this example, RE1 exceeds the threshold by the largest amount, indicative of a Lane I (sensors 1 & 4) failure, and Triad 3 data (computed using Lane II sensors 2 & 5 and Lane III sensors 3 & 6) is selected. If a sensor or lane failure source goes away, time based hysteresis enables the sensor or lane to be reinstated; the failure detection variables eventually drop below the threshold, and the average of the three triads is again used.

Alignment Bias Updating

During pre-launch alignment, the system bypasses the sensor redundancy logic and defines sensor data as the average of the three triads. System bias values are defined from this data which make each of the gyro triads' and the accelerometer triads' horizontal axes equal to their respective average triad data. The X (vertical) axes of the accelerometer triads' data are biased to match the known gravity. Also, a gyro system bias is applied to the average gyro triad data to make it reflect known earth rate components. These bias updates, in addition to making the system reflect known earth rate and gravity conditions, will make the three triads' redundancy data exactly equal at the time of launch.

Sensor performance problems are reflected in the magnitude of the system biases used to match earth rate and gravity. Isolation of the problem to a given lane will be reflected in the bias values used to match triad data to the average triad values. Nominal alignment in the X up position detects all accelerometer performance problems because all accelerometers sense a component of vertical acceleration.

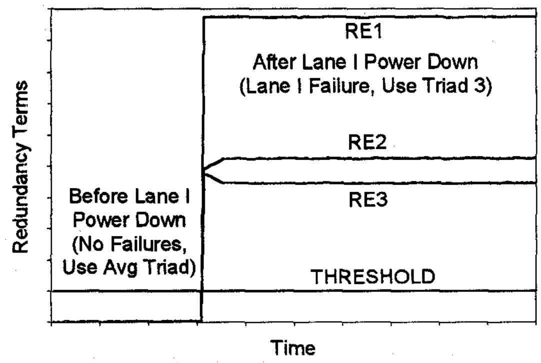

Figure 5: Gyro Sensor Redundancy Example – RIFCA S/N 134 Lane I Power Down Test

Computational Redundancy and Output Data Selection

Output data selection is restricted to data from Lanes I or II to facilitate computational redundancy with a simple two out of three voting technique. Lane III data is used only to aid in Lane I or II selection. Data voting is done by each of the processors independently, such that each processor votes for Lane I or Lane II data to be transmitted via the 1553 data buses. This vote is done by comparing key data between lanes and with the assumption that if two lanes' data compare within an acceptable tolerance, both lanes are good. The order of comparison and vote selection is such that priority is given to the lane whose output was selected on the last cycle. If no lanes have comparable data, the vote is given to the lane used for output on the last cycle, so that there will always be a vote selection by each processor.

SOFTWARE INTEGRATION

The flight software is integrated as a real time interrupt driven package. There are three primary computation frequencies all driven from a hardware generated 400 Hz interrupt. The primary function of the 400 Hz interrupt is to do RIFCA inertial sensor data processing and count down for a software generated 50 Hz interrupt. The 50 Hz interrupt performs system level sensor data processing, attitude navigation, controls, vehicle discrete and controls data output voting, sequencer time, telemetry data collection and output, 50 Hz to 2 Hz interface data processing, 2 Hz to 50 Hz data interfacing, and in pre-launch, alignment and vehicle test functions. Like the 400 Hz processing, the 50 Hz processing counts down for a software generated 2 Hz interrupt. The 2 Hz interrupt supports position and velocity navigation, guidance, updating of slowly varying control gains and redundancy limits, and the low frequency functions of load relief.

In addition to the above real time computations, a cyclic background task is implemented where non-time critical functions are performed. These functions include memory checksum, A/D temperature and health monitoring, sensor temperature compensation updates, and pre-launch control of memory monitoring and updating, and mode control.

In a pre-launch configuration the program can be executing in three primary modes: (1) Idle, during which real time interrupts are disabled; (2) Alignment, during which all but the 2 Hz interrupts are active; and (3) SIM Flight, during which the system is operating in a flight mode but augmenting the sensor data from a pre-launch active SIM flight vehicle model. The SIM flight model is driven from the flight program data of control and sequencing signals and provides the expected system level sensor data resulting from these signals using a normal vehicle transfer function.

In addition to the real time operation indicated above, there are several interrupts supporting asynchronous functions of: (1) Ground message handling (RS 422 up and down interrupts); (2) Output data selection implementation (Data I or Data II interrupts); (3) Asynchronous sequencing of second engine cut-off commands (Timer A interrupt); and (4) The 1553 bus data complete interrupt.

DATA SHARING

Data is shared between the three processors via three dual port RAMs, one in each lane. In general, shared data is placed in these RAMs when it is computed. The data is read at specific locations in the code where timing analysis ensures data availability under worst-case timing and alternate path possibilities. A failure of one of these RAMs will still allow two of the processors to communicate, since each dual port RAM only supports the data transfer between two processors, which is all that is required for mission success.

SENSOR TEMPERATURE COMPENSATION

There are six temperature sensors in each lane associated with the gyros, accelerometers and V/F converters. These sensors are read in the software background loop approximately once every second. In this loop, updated calibration coefficients for gyro bias and accelerometer bias and scale factor are computed. Thermal compensation for the V/F converters is included as part of the accelerometer scale factor and bias compensation. The gyro and accelerometer thermal sensors are all in the ISA and have similar temperature profiles. To take advantage of this and provide a degree of robustness for a potential thermal sensor failure, an average ISA lane temperature profile is defined as the average of the four thermal sensor readings. Each thermal sensor also maintains knowledge of its bias from this average temperature. Should a thermal sensor fail, its reading will be replaced by the average temperature profile plus its last computed bias. The average profile will continue to be computed using all sensors, including the computed value for the thermal sensor that failed.

SYSTEM PERFORMANCE

Each RIFCA unit is thoroughly tested at both the component and system levels during the build process and acceptance testing. Many of the key performance parameters are tracked and maintained in databases, including system level error budget terms such as in-run and long term gyro and accelerometer performance characteristics, and short term system level calibration coefficient stability. The databases currently include data from 35 production units, and provide valuable performance related statistics. Current RIFCA performance capability based on these databases is summarized in Table 3.

Table 3: RIFCA Performance Capability

Error Source	Total Error (3-sigma or max)
System Gyro	
Bias	0.05 deg/hr
Scale Factor	10 ppm
Misalignment	10 arcsec
Random Walk	0.007 deg/rt-hr
Quantization	0.65 arcsec/pulse
System Accelerometer	
Bias	150 μg
Scale Factor	250 ppm
Misalignment	15 arcsec
K2 Scale Factor	20 $\mu g/g^2$
Quantization	0.01 ft/sec/pulse

SIMULATION AND ANALYSIS CAPABILITY

AlliedSignal has developed various analysis tools for RIFCA, including a generalized C-language covariance simulator used for orbital mission level error budget analysis, and a detailed RIFCA mechanization simulator which consists of a C-language driver that calls the strapdown and navigation Ada op-code. The mechanization simulator was designed to simulate static and dynamic navigation/redundancy lab test scenarios and has a record of accurately imitating lab test results when lab test conditions are known. These analysis tools are periodically upgraded and enhanced as more RIFCA system experience is gained.

PRODUCTION FLOW

The system level production flow for each Boeing Delta II RIFCA flight unit consists of 5 major steps: (1) Acceptance Test, (2) Receiving Test, (3) In-Vehicle Test, (4) On-Stand Test, and (5) Pre-Launch and Flight.

The first step is a thorough 1-month acceptance test, which is performed at AlliedSignal after each unit is integrated. The system is exposed to calibration, alignment and navigation test profiles conducted under ambient, thermal, vibration and vacuum environmental conditions to assess system performance. Upon completion of the acceptance test procedure, the unit is shipped to Boeing Delta Mission Check Out (DMCO) at Cape Canaveral where it undergoes a 1-day receiving test. The receiving test includes system performance, alignment, navigation, static, dynamic, lane down, and porro prism testing. The unit is then placed in storage, typically for 3 to 4 months.

The next step in the production flow is a 2-week, second stage in-vehicle test, which occurs at Boeing DMCO. This test includes system performance, alignment, vehicle compatibility, vehicle sequencing, controls output, and simulated flight tests. The unit is then placed in storage again, typically for another 3 to 4 months.

The fourth step, a 3 week test conducted on a Boeing test stand, includes system performance, alignment, vehicle and mission compatibility, vehicle sequencing, controls output, and simulated flight tests. The final step in the RIFCA production flow consists of pre-launch testing on the launch pad and the flight itself, with a combined time of approximately 3 to 4 hours. Pre-launch testing includes system performance and alignment tests, and the flight yields flight accuracy, post flight analysis, and mission success and anomaly data.

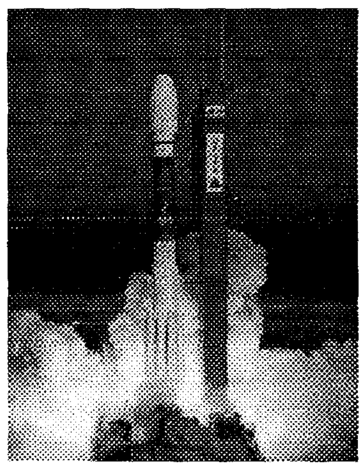

Figure 6: A RIFCA Guided Boeing Delta II Launch

FLIGHT RESULTS

At the time of this writing (February 1998), there have been 17 RIFCA guided Boeing Delta II launches. A typical launch is shown in Figure 6. Orbital insertion accuracy requirements were easily met for each successful mission, as summarized in Table 4. Some noteworthy facts regarding RIFCA performance during these missions follow:

- RIFCA S/N 102 was launched 19 months after final system level factory calibration at AlliedSignal, which is indicative of excellent long-term stability of the system.

- RIFCA performance is best observed during the seven two-stage missions highlighted in Table 4. The third stage of each three-stage mission is unguided and contributes most of the vehicle orbital insertion accuracy errors.

- During the recent SKYNET-4D post mission depletion burn, an engine anomaly occurred causing excessive vehicle rates up to 250 deg/sec, which were sensed by RIFCA S/N 119. This anomalous condition caused several triggerings (as expected) of gyro redundancy, selecting different computed body frame triads, but always providing continuous good quality rate data. In addition, a one time triggering of accelerometer redundancy due to excessive tangential accelerations also resulted in smooth continuous acceleration data to the navigation system. The tangential acceleration problem was corrected and after the specified holdout interval, all accelerometers were again used in defining the acceleration data.

- Finally, RIFCA S/N 110 operated properly during the GPS IIR-1 mission, sensing anomalous rates and accelerations up to vehicle destruction. The unit was recovered afterwards, and although sustaining some housing damage, its condition was good enough to warrant repair as an engineering unit.

PLANNED RIFCA IMPROVEMENTS

The existing RIFCA design currently meets all Boeing Delta II mission requirements; however, AlliedSignal's commitment to continuous improvement and the fact that there are tighter Delta III and Delta IV requirements to be met in the future, has led to the development of a plan to improve RIFCA's performance, reliability and cost. Now that AlliedSignal has built, tested and shipped 35 RIFCA flight units, 17 of which have flown on Boeing Delta II launch vehicles, specific aspects of the system have been defined as opportunities for improvement. Planned hardware, software and system level modifications are expected to yield the desired improvement in the RIFCA product.

Hardware modifications include upgrading the accelerometer to the QA3000 Isocoil version built by AlliedSignal Instrument Systems, increasing the robustness of the V/F low pass filter design to component variation, and increasing processor throughput and memory capacity. Software modifications include a new gyro resynchronization algorithm, implementing the sensor redundancy algorithms in the sensor frame instead of the body frame, thus allowing for operation in the X-horizontal orientation, and new calibration equations which are insensitive to rate table sag errors. System level modifications include an improved calibration process, especially for the accelerometer and V/F channel, optimizing calibration data sampling times, and determining the exact sources of vibration induced attitude and velocity errors (coning and sculling).

SUMMARY

AlliedSignal's RIFCA has established itself as a reliable flight control assembly for the Boeing Delta II launch vehicle, based on a history of 35 flight production units being integrated, tested and shipped, and 17 units flown. With continuous improvements based on years of experience from the initial concept through the latest launch, RIFCA is expected to continue to be a practical, cost-effective, fully fault tolerant inertial system for many years to come.

ACKNOWLEDGEMENTS

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Table 4: RIFCA Guided Boeing DELTA II Launch Vehicle Orbital Insertion Accuracy

Payload	Launch Date	RIFCA S/N	Apogee (nmi)		Perigee (nmi)		Inclination (deg)		RAAN (deg)		Perigee Velocity (fps)		Arg of Perigee (deg)		C3 (Km2/Sec2)		Geodetic Lat (deg)	
			Req	Act	Req	Act	Req	Act	Req	Act	Req	Act	Req	Act	Req	Act	Req	Act
*XTE	12/30/95	101	+3.1	+1.1	+3.1	+0.1	±0.01	0.00										
GPS-10	7/16/96	104	±225	-4	≥85.0	+0.9	±0.50	0.06	±0.50	-0.03								
GPS-8	9/12/96	108	±225	-31	≥85.0	0.6	±0.50	0.06	±0.50	0.04								
MGS	11/7/96	103			+3.87	+0.06	-0.09	0.00			+28.7	-11.3	+0.69	+0.18	+0.18	+0.08		
MPF	12/4/96	111			+2.36	+0.15	+0.31	+0.08			+34.9	+10.8	-0.63	-0.11	+0.23	+0.08		
**GPS IIR-1	1/17/97	110	N/A		N/A		N/A		N/A		N/A		N/A		N/A		N/A	
*MS-1A	5/5/97	107	+5.0	+0.7	-5.0	-0.6	±0.04	0.00										
Thor II	5/20/97	116	-1261	-152	3.1	0.3	-0.28	-0.07					0.67	0.02				
*MS-2	7/9/97	114	±5.0	+1.1	±5.0	+0.2	±0.04	0.00	+0.06 -0.04	0.00								
GPS IIR-2	7/23/97	109	±210	+21	±4.0	+0.8	±0.40	0.00	±0.20	0.02							±0.60	+0.08
*MS-3	8/20/97	112	±5.0	+1.3	±5.0	+0.3	±0.04	0.00	+0.06 -0.04	0.00								
ACE	8/25/97	***102	+4.0	+1.0			±0.03	0.00			-27.0	-4.2	±0.09	0.00				
*MS-4	9/27/97	105	±5.0	+2.6	±5.0	+0.2	±0.04	0.00	+0.06 -0.04	0.00								
*MS-5	10/31/97	124	±5.0	+1.4	±5.0	-0.4	±0.04	0.00	+0.06 -0.04	0.00								
GPS-9	11/6/97	122	±225	+37	≥85.0	-0.1	±0.50	0.00	±0.50	-0.02								
*MS-6	12/20/97	125	±5.0	+0.4	±5.0	+0.1	±0.04	0.00	+0.06 -0.04	+0.01								
SKYNET-4D	1/10/97	119	+452	+199	±2.9	0.0	±0.24	+0.03					-0.59	-0.02				

* Two Stage Missions   ** Vehicle Failed Shortly after Lift Off   *** This Unit was 19 Months from Calibration When Flown